Assignment-03

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Section: 19

CSE340

Ans. to the ques. no. 1

In IEEE-754, the exponent is stored as an unsigned integer using a biased representation rather than 2's complement. It uses a bias so that all stored exponent become non-negative.

Formula: Biased exponent = Actual Exponent + Bias

$$\left.\begin{array}{l} \text{If, } +5 = 127 + 5 = 132 \\ 0 = 127 + 0 = 127 \\ -5 = 127 - 5 = 122 \end{array}\right\} \text{Stored exponent (Single)}$$

Here, for positive exponent, stored value is greater than bias while in negative exponent, stored value is lesser than bias. The stored value is always non-negative, enabling simple unsigned comparisons and hardware design.

Ans. to the ques. no. 2

In long multiplication, there are many shift + add operations, which makes the time complexity to be $O(n^2)$. While on the other hand, optimized multiplication reduces the number of operations, handles sequences efficiently and does parallel processing. This makes their time complexity to become $O(n \log n)$ or even better. The optimized multiplication method eliminates the need to store and align multiple partial products, reducing both time and space overhead significantly.

Ans. to the ques. no. 3

Iteration	Multiplicand	Product
0	10110	00000 101
1	—	10110 101
	—	01011 010
2	—	—
	—	00101 101
3	—	11011 101
	—	01101 110

Ans. to the ques. no 4

Fraction $= 14$ bit

Sign $= 1$ bit

Exponent $= 24-(14+1) = 9$ bits

Here,

$$\left((-0.0101)_2 \times 2^{79} \quad \times \quad (10010.0101)_2 \times 2^{87}\right)$$

$$= (-1.01) \times 2^{79-2} \times (1.00100101) \times 2^{87+4}$$

$$= (-1.01) \times (1.00100101) \times 2^{168}$$

$$= (1.01101110010000)_2 \times 2^{168}$$

Actual exponent $= 168$

Bias $= 2^{9-1} - 1 = 255$

$\therefore$ Biased exponent $= 168 + 255$

$= 423$

Biased exponent range $= 0$ to $2^9 - 1$

$= 0$ to 511

$= 1$ to 510

That is,

$1 < 423 < 510$ which means no overflow or underflow.

Ans. to the ques. no. 5

```
flw f14, 0(X13)

flw f7, 4(X13)

addi X10, X0, 0        // X10 = c = 0

flt.s X11, f14, f7

beq X11, X0, else

fcvt.ws X10, f14

jal X0, Exit

Else:

    fcvt.ws X10, f7

Exit
```

fp = array [0] = X_{13}

a = f_{14}

b = ~~f14~~ f_7